

DIPAYAN SARDAR

CIVIL ENGINEERING & DATA SCIENCE UNDERGRADUATE

Kolkata, India | +91 9733817692 | dipayansardar73@gmail.com

LinkedIn: [linkedin.com/in/dipayan-sardar-a5119a381/](https://www.linkedin.com/in/dipayan-sardar-a5119a381/) | **GitHub:** github.com/dipayansardar73-decode/ |

Portfolio: portfolio-frontend-1n80.onrender.com | **Twitter:** x.com/dipayansardar73

SUMMARY

Interdisciplinary **Civil Engineering** and **Data Science** undergraduate with strong expertise in AI/ML, Data Science, and engineering research. **Global Top 1000 in AWS Aldeas 2025**. Research experience in fluid mechanics and environmental systems with AI-driven modeling and optimization. Experienced in full-stack development and scalable technical solutions. Focused on building scalable AI-driven engineering and software solutions in research and industry environments.

EDUCATION

Bachelor of Science, Data Science

IIT Madras

Expected 2028

- Coursework: Python, Data Science, Deep Learning Fundamentals

Bachelor of Engineering, Civil Engineering

Expected 2028

Jadavpur University, Kolkata

- Coursework: Structural Analysis, Water Resources, Fluid Mechanics, Environmental Engineering

AWS Aldeas Competition 2025 – Top 1000 Worldwide

TECHNICAL SKILLS

Languages : Python, javascript

Data & Streaming : PostgreSQL

Frontend & Mobile : React, HTML5/CSS3, Tailwind

Backend : Node.js

AI/ML : PyTorch

Tools : Git, VS Code, Jupyter, Figma, Canva

Engineering: AutoCAD

PROFESSIONAL EXPERIENCE

Research Intern at IIT Bombay

May 2026

- Soil Erosion Control using Flume Experiment

Member – American Society of Civil Engineers (ASCE)

Jan 2024 – Present

- Participate in technical symposiums and infrastructure-related events

Core Coordinator – Jadavpur University Entrepreneurship Cell

Jan 2024 – Present

- Coordinated innovation initiatives and entrepreneurship programs
- Organized **Hult Prize** Kolkata round, **JU E-Summit**, **Srijan Tech Fest**

Core Coordinator – JU Association for Computing Machinery (ACM) Student Chapter

Jan 2024 – Present

- Organized technical workshops on AI/ML applications and coding competitions

Event Manager – Srijan Tech Fest

- Served as Point of Contact for events with 2000+ participants

UI-UX and Professional Graphics Designer

- Successfully managed and delivered numerous premium-scale projects for many Organisations

Open Source contributor

- Like to contribute Open Source Projects, have done contributions in Org like IfcOpenShell, gprMax

RESEARCH EXPERIENCE

Student Researcher (Environmental Engineering) – Jadavpur University

Jan 2025 – Present

- Conducting research in environmental engineering with focus on **wastewater management** and treatment systems
- Working in environmental engineering labs on water quality analysis and treatment processes
- Studying pollutant removal mechanisms and optimizing treatment efficiency
- Analyzing experimental data to evaluate treatment performance

Independent Researcher (Fluid Mechanics) – Jadavpur University

Jan 2025 – Present

- Conducting independent research on **water leakage management systems** in distribution networks
- Studying flow dynamics, pressure variations, and leakage detection methodologies
- Developing experimental setups to analyze leakage patterns in pipeline systems
- Investigating hydraulic parameters affecting water loss in urban infrastructure

PROJECTS

Bhumi – AI Platform for Farmers | React, Azure Speech Services, IoT

- Multilingual platform (9+ languages) for crop prediction, disease detection, and irrigation advice
- Voice-enabled assistant (Bhumi Live) piloted with local farmers

Vastu 2.0 – Building Design & Analysis | Computational Geometry, Python, AI-ML, Three.js

- Automates structural design and analysis for civil engineering applications

Automated Railway System Optimization | Python, Optimization Algorithms

- Designed scheduling optimization models for railway systems
- Applicable to highway traffic management

Water Leakage Detection System | Python, ML, Sensors

- Built IoT + ML solution for pipeline leak detection and localization
- Reduced water loss by 30% and maintenance response time by 40%

ACHIEVEMENTS

AWS Aldeas Competition 2025 – Top 1000 Worldwide

Winner of HackNpitch Hackathon

3rd intercollege round of SIH

LAUNGUAGES

- English – Fluent
- Hindi – Fluent
- Bengali – Native

PRs

<https://github.com/SandeepVashishtha/Eventra/pull/987>

<https://github.com/RatLoopz/sahidawa-india/pull/55>

<https://github.com/SdSarthak/AegisAI/pull/216>

<https://github.com/Muneerali199/Draftdeckai/pull/439>

LIVE PROTOTYPE

<https://vastu-v3.vercel.app>

PROJECTS LINK

<https://github.com/dipayansardar73-decode/Vastu-V3>

<https://github.com/dipayansardar73-decode/VASTU-X2-VERSION-2.0-.git>

https://github.com/dipayansardar73-decode/BHUMI_THE_FUTURE_OF_AGRICULTURE

<https://github.com/dipayansardar73-decode/gig-verification-portal>

DESIGN PROJECTS

<https://canva.link/png2b0f0xus7l4r>
